I. Research Question

Why is our perception of knowledge conflated to ideas of fact and empiricism, and how has this been shaped over the course of philosophical discourse?

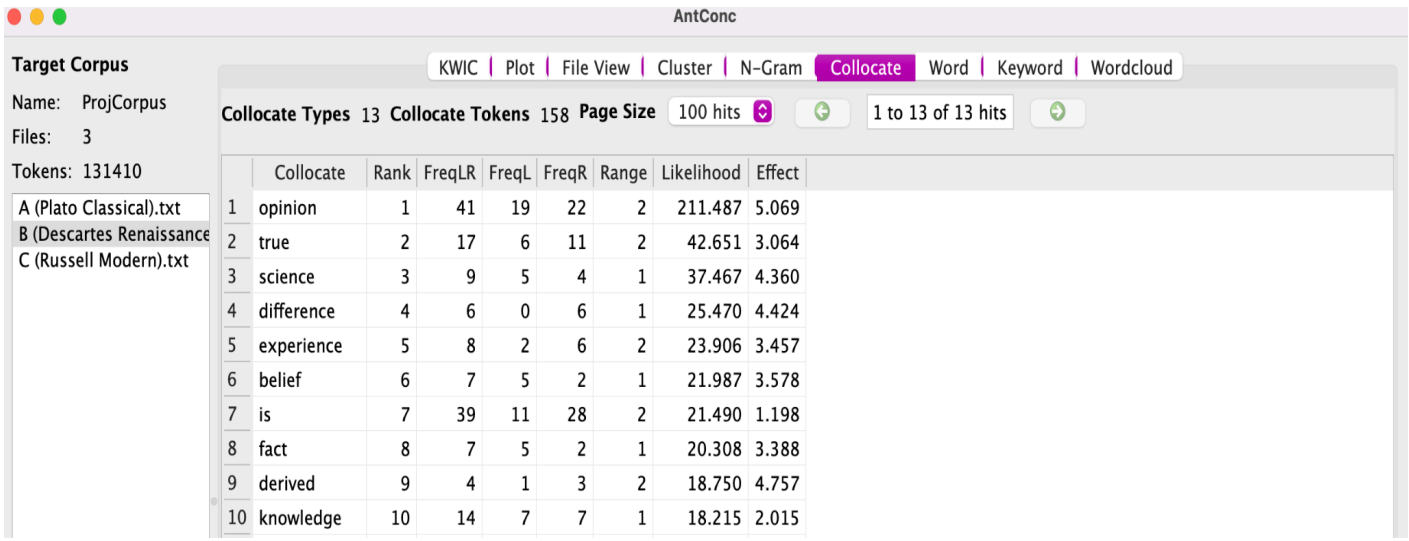
We will investigate this question by looking at three texts focused on uncovering knowledge through the philosophical lens. We created a relatively small corpus of texts that included: *Theaetetus* by Plato, *Meditations on First Philosophy* by René Descartes, and *The Problems of Philosophy* by Bertrand Russell. The ideas in this text span classical philosophy, renaissance philosophy, and modern philosophy, respectively. At these three distinct points, representative of the thinking at the time, we hope to gain insight into how knowledge is perceived and related throughout our societal evolution. The question highlighted above is a means to guide our thinking and computational investigation. We present our insights below.

II. Data Analysis and Visualizations

Visualization 1:

Row	FileID	FilePath	FileTokens	Freq	NormFreq	Dispersion	Plot
1	2	C (Russell Modern).txt	43127	307	7118.510	0.776	
2	0	A (Plato Classical).txt	66149	308	4656.155	0.782	
3	1	B (Descartes Renaissance).txt	22134	17	768.049	0.784	

Figure 1.1 is an AntConc visualization depicting a normalized frequency plot of the term "knowledge" across the texts in the corpus.



	Collocate	Rank	FreqLR	FreqL	FreqR	Range	Likelihood	Effect
1	opinion	1	41	19	22	2	211.487	5.069
2	true	2	17	6	11	2	42.651	3.064
3	science	3	9	5	4	1	37.467	4.360
4	difference	4	6	0	6	1	25.470	4.424
5	experience	5	8	2	6	2	23.906	3.457
6	belief	6	7	5	2	1	21.987	3.578
7	is	7	39	11	28	2	21.490	1.198
8	fact	8	7	5	2	1	20.308	3.388
9	derived	9	4	1	3	2	18.750	4.757
10	knowledge	10	14	7	7	1	18.215	2.015

Figure 1.2 is an AntConc visualization showing the top 10 collocate terms of "knowledge" across the texts in the corpus

	Cluster	Rank	Freq	Range
1	knowledge is	1	16	2
2	knowledge of	2	12	2
3	knowledge as	3	4	1
4	knowledge and	4	3	2
5	knowledge but	4	3	1
6	knowledge or	4	3	1
7	knowledge when	4	3	2
8	knowledge they	8	2	1
9	knowledge what	8	2	1
10	knowledge are	10	1	1

Figure 1.3 is an AntConc cluster visualization showing the top 10 phrase clusters that appear with the term "knowledge".

Explanation 1:

The first data visualization was conducted in AntConc, it explores a couple of useful things, helping us get a sense of what knowledge is from a distant reading perspective. If you look at Figure 1.3 you will see that the most frequently appeared clusters are the target term followed by various types of grammatical structures. One might find that this is too broad to make some sort of idea about, so we selected the top 2 clusters to center our thoughts around. These clusters: “knowledge is” and “knowledge of” are indicative of authors making statements describing or explaining knowledge. The preposition relates a noun to another word and the verb expresses some action or state of being. We take from this that the authors of at least 2 of the texts sought to make descriptive or explanatory statements about knowledge. Why is this useful to know? Our investigation relies on there being some preconceived notion about knowledge and this information is a good indication that such notions exist.

Figure 1.2 identifies the most frequent collocations that appear within 5 words to the left or right of the target term. When paired with the clusters, one can clearly see how such words can be made descriptors of knowledge: what it is or what it does. Interestingly enough the most frequent collocation is not a word we would readily associate with the target term. What does opinion have to do with knowledge? Interestingly enough, if given some thought you may realize that a lot of what we consider knowledge started out as being someone's opinion. And most of the time those opinions were unpopular, take for example the knowledge that we live in a heliocentric orbit rather than geocentric. This was quite an unpopular opinion, but it was later proven to be true then taken as a fact and ultimately knowledge about the way that we exist. Is it that all knowledge starts out as opinion? We’ll leave that as an exercise to the reader.

Lastly, Figure 1.1 describes the dispersion of the target term across the texts in the corpus. It identifies a frequency of how many times that term appears in a specific text and ranks them. One can see from the plot visualization that modern and classical periods make broader use of the term compared to the renaissance period. We will talk about this more in the next section. The real intent of this plot was to be a dummy check, more or less, we wanted to ensure that we did pick texts that did make mention of the target word. More specifically than that, the target word needed to have appeared in such a fashion that one could say it was a key topic in each text. We must also point out here, that one could have similarly engaged the same verification utilizing topic modeling instead.

All in all we provide this visualization and analysis to show the general inferences and observations that can be made reading distantly or when the scale of inquiry renders other options unavailable.

Visualization 2:

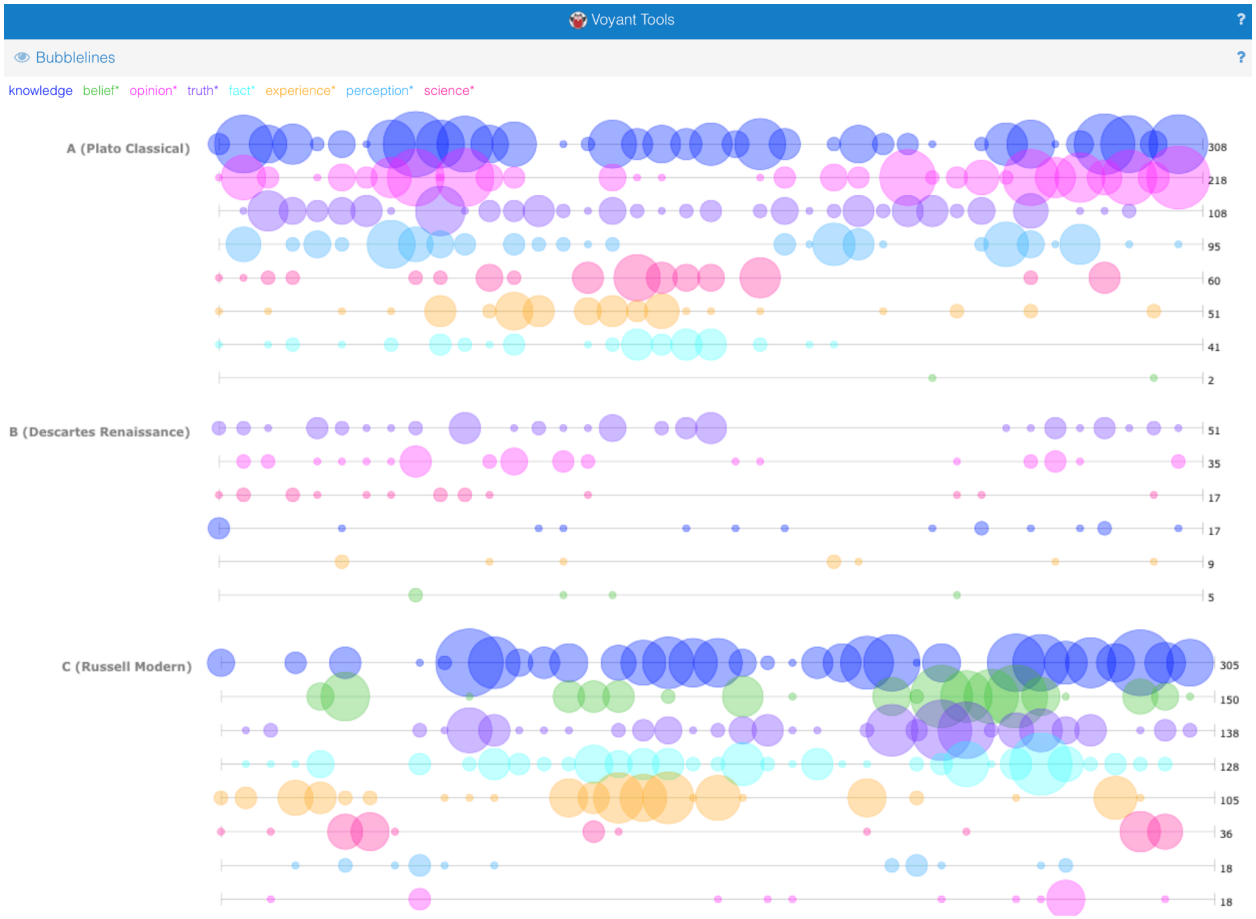


Figure 2 is a Voyant bubblelines depiction showing both the density and spread of terms across each text in the corpus.

Explanation 2:

The second data visualization looks more closely at each of these texts, trying to sort out differences between how knowledge is understood in each of these philosophical contexts. The bubblelines tool looks at both the density (how much does a term appear in an instance) and the spread (where do the terms appear). Looking at Plato, we see that when knowledge is spoken about (very frequently) it seems to be spoken about with regards to opinion and truth. We infer from this

that the classical philosophy regarding knowledge is more about sharing opinion and truth as a form of knowledge rather than data or fact.

For Descartes, we find that none of the terms we are looking for show up with any significance. This is perplexing given that this philosophical period is around the time of forming the foundations for scientific inquiry and empirical methodology. Thus a lack of engagement with the terms presented is puzzling, even more so knowing that *Meditation on First Philosophy* is known for its groundwork on establishing certainty in knowledge as a foundation for the sciences.

This is a perfect place to point out some of the differences between performing a close reading and a distant reading. Our knowledge of Descartes text in a close reading form provides me with the information that explains and describes what the text is about. In other words, close reading is the only way that we know at all that Descartes' text even engages a philosophical framework on knowledge. This could not be gotten by distant reading. As we pointed out previously, the visualization might lead one to conclude that we had selected the wrong text for this corpus (we are still not entirely sure that this isn't true). To be clear here: the issue that we are seeing with target terms, their density, and their spread, is an issue of how much Descartes spends time talking about these terms. Most of his book is a discussion on mind, body and G-d, thus his focus is not on the topic of knowledge in the way that it is for the other two others of the corpus. This doesn't negate the value of his book on Renaissance philosophy of knowledge, it is regarded as one of the most prominent thoughts of the time. The focus is just elsewhere; information that is not so clear when reading distantly.

Lastly, for Russell emphasis is placed on belief, fact, and truth. It seems here that claims about knowledge are related not to empirical fact as data but rather some abstract concept of faith. This could suggest that in the modern period, philosophers see knowledge as coming from within, an idea of innate understanding or commitment to faith. Both ideas are quite intriguing and they relate well to our previous mentioning of opinion as knowledge from visualization 1. This would just be a different flavor of the same idea, after all isn't this what we expect to see in the knowledge production process. As an aside, a math professor once mentioned to his class that "mathematics is 10% fact and 90% faith" knowing what you know now, would you agree?

Visualization 3:

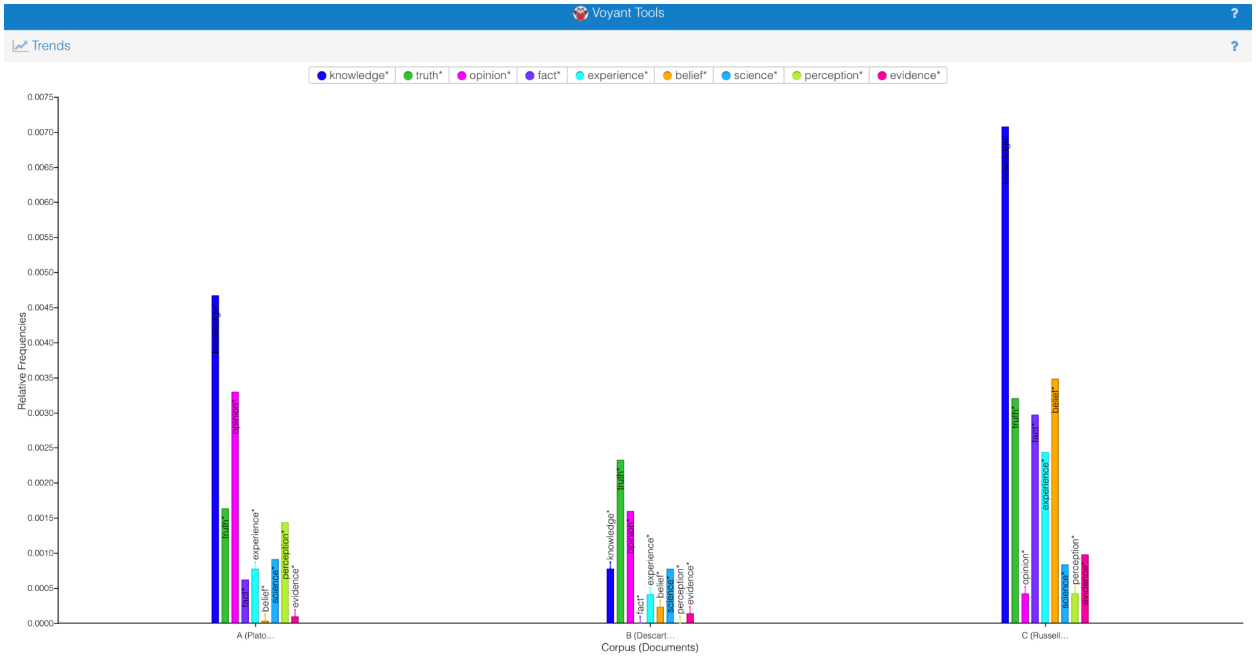


Figure 3 is a Voyant depiction of the target word trends across the corpus.

Explanation 3:

The final visualization is trend showing the relative frequency of the target terms. Based on our previous exploration of the Descartes work we would do well to read these results in context. Although across the philosophical discourse it appears as though the only periods concerned with knowledge are the classical and modern periods, however we know from a close reading perspective that the renaissance period is also quite invested in the topic. Thus reading the results in context means that we can make some cautious observations. We do that here by looking at the relative frequency of the target term truth.

The relative frequency of truth appears to increase across the discourse, this says that even when philosophers are not particularly focused on understanding knowledge through the same conventions they still particularly value the concept of truth. Now, what does truth have to do with knowledge? This is a question that exists outside of the scope of this research, however it is quite clear that when we speak of knowledge we find that talking about truth is correspondingly

important. I think that this is particularly fitting because it allows us to generalize to the present.

In our current climate we find ourselves concerned with mis- and dis-information. The question about what is true or false is becoming increasingly hard to answer, but it's curious to see that there is a historical trend of talking about the truth. In context this trend parallels technological innovations, culture shifts, and societal change, can we then say something about truth as we continue to develop with time? Short answer: probably. But to take this back into the direction of knowledge, we are coerced to claim that as we advance and grow our societies, what we think we know becomes so fraught and different that we attempt to root our knowledge in truth.

We are curious to see, now, if we were to investigate some of the more recent philosophical discourses on knowledge, what trends would we see? Making an educated guess, truth must almost certainly appear to increase in frequency. Maybe given our global political climate that issues of knowledge have morphed into investigations on truth. Nevertheless, these historical trends show, importantly, changes in thought. Which at the very least is a promising sign that we haven't always thought about things in the same way, in other words we are changing and learning.

What we concluded from our analysis is that according to the corpus, historically knowledge has not been conflated to ideas of fact and empiricism; in fact what we are seeing is that ideas of knowledge have always been debated. We even did this during the semester, attempting to determine what different types of reading do to knowledge production. We had to come up and defend what knowledge was to us. So no knowledge is not conflated in any direction, we are still trying to piece together what it is. The task is so difficult that it seems one of the only ways to do it is to look at it in terms of other concepts and try to make sense of it from them.

III. Methods

To do this analysis we used OpenRefine, AntConc, and Voyant tools. After forming a research question we began looking for sources using Project Gutenberg, of which we found three. Our search for sources was contained to finding one text that represented a specific philosophical period and pertained to some interrogation of knowledge. Once these sources were collected we then used OpenRefine to clean our data (corpus). Adhering to the common

transforms, each source was cleaned and re-downloaded into the corpus file. We then took the corpus file and uploaded it to AntConc to perform our analyses. All AntConc analyses utilized advanced search. The target terms are those labeled at the top of Figure 2 and Figure 3. Subsequent to that, we then conducted the final analyses on Voyant tools, using the same target terms as in the AntConc analyses, taking screen snips and presenting them here.

IV. Concept analysis

Although we did conduct a computational analysis, we want to reflect on the literary concepts that got us here. We want to particularly address concepts of scale. As we pointed out at different points in our analysis, looking at the data from varying scales afforded us certain capabilities and ultimately information. The concept of scale talks about things in terms of physical size but also more abstract parameters like say a level of understanding. What can be gotten from different scales is of varying value as we tried to make apparent in our own analysis, scaling between close and distant reading was so valuable that it could be described as: necessary. Moving freely between scales is arguably fundamental to knowledge production, so it is fitting for an inquiry into knowledge that we make sure to manipulate scale as we have done here.